

NORAIZ RANA

Task #24

Topic

Weather App in React
using OpenWeatherMap Api

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SUBMITTED TO:

APPVERSE TECHNOLOGIES

ClimaPulse – Project Report

Introduction

The ClimaPulse is a web application developed using modern web technologies to provide real-time weather updates. The app fetches data from the **OpenWeatherMap API** and displays information such as temperature, weather conditions, humidity, and wind speed for any searched city. The primary goal of this project is to practice API integration, environment variable handling, and front-end development using React, Vite, and Tailwind CSS.

Weather forecasting plays an important role in daily life. People need quick and reliable weather information for planning outdoor activities, travel, and work schedules. This project focuses on delivering weather data in a **user-friendly and responsive interface** with features like dynamic search, dark mode, and smooth animations.

Technical Breakdown

The app is built on the **React + Vite** framework, which ensures fast development and efficient rendering. The styling is done using **Tailwind CSS**, which provides utility-first classes and a clean design system.

Key features include:

1. API Integration

- The app uses the **OpenWeatherMap REST API** to fetch weather details.
- The API key is stored in a `.env` file for security and is accessed in the code using

```
import.meta.env.VITE_OPENWEATHER_API_KEY.
```

2. Search Functionality

- Users can enter the name of a city in the search bar.
- The app fetches and displays live weather information for the requested city.

3. Weather Display

- Shows temperature, weather condition (clear, rain, snow, etc.), humidity, and wind speed.
- Icons and text update dynamically based on API response.

4. Responsive Design

- Tailwind CSS ensures the app looks good on desktop, tablet, and mobile devices.
- Layout adapts automatically to screen size.

5. Dark Mode and Animations

- Implemented dark/light mode toggle for better user experience.
- Smooth transitions and hover effects enhance interactivity.

6. Error Handling

- Displays clear messages when the API limit is reached or the city name is invalid.
- Handles cases where API key is missing or incorrect.

Challenges Faced

1. API Key Handling

- Initially, the API returned an “Invalid API Key” error.

- This was solved by checking OpenWeatherMap account settings and placing the key correctly in the `.env` file with the `VITE_` prefix.

2. Cross-Origin Requests (CORS)

- Sometimes the browser blocked requests during local development.
- Running the app with Vite's development server helped bypass this issue.

3. UI Styling

- Creating a clean and modern interface using Tailwind CSS required experimenting with layout, spacing, and colors.

4. API Limitations

- The free OpenWeatherMap API has a limit on the number of calls per minute.
- To handle this, proper error messages were added to guide the user.

Learning Outcomes

From this project, I learned:

- How to fetch data from a REST API and display it dynamically in a React app.
- How to securely store and use environment variables in a Vite project.
- How to use Tailwind CSS for building responsive and attractive user interfaces.
- How to manage errors and edge cases while working with APIs.
- Importance of debugging and reading API documentation carefully.